The chain

01 Chip Design, EDA and IP
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Chapter 12 Stage 12 of 14 High concentration · Moderately hard to substitute

Test and Assembly

Testing finds bad chips before they go into an expensive package. Japan's Advantest took about 65 percent of 2025 spending on chip testers, by its own estimate.

Specimen 12 · Test and Assembly

A vertical probe card over a wafer. It is called vertical because its needles point straight down at the wafer. The card is round, about 305 mm across, a little wider than the 300 mm wafer beneath it. Thousands of needles under it land on four dies at once, and those four dies are picked out in blue-violet. The gap between card and wafer is exaggerated. The real gap is a millimeter or two.

In plain terms

Before a chip can be sold, a machine called a tester has to check that it works. While the chips are still on the wafer, thousands of needles press onto small metal pads on each one. The tester sends signals through the needles and reads what comes back, and any chip that fails goes no further. A modern AI accelerator is a dozen expensive pieces bonded into one package, and if one piece is bad, all of them are thrown away. So the industry now tests chips earlier, at higher temperatures and for longer, to make weak ones fail during testing before they reach a finished product. Two firms, Advantest in Japan and Teradyne in the United States, sell most of the testers.

In short

In a TrendForce ranking cited by CSET, four of the ten largest chip assembly and test contractors by 2024 revenue are headquartered in China[1]. Georgetown's CSET, analyzing TechInsights data, finds that between 2019 and 2024 Chinese toolmakers gained 5 percentage points of world market share in test tools[1]. Across assembly, test and packaging tools, they still lag significantly behind foreign suppliers in market share[1]. Advantest says that switching to another tester maker means rebuilding a customer's whole test setup, at significant extra cost[2].

Turning silicon into a finished, packaged chip takes 1,000 to 1,500 steps, depending on how complex the chip is, the Semiconductor Industry Association estimated in 2025[3]. Of all those steps, only test puts electrical signals through the chip, Advantest's investor guide of July 2026 says[4]. The

1 **Inside Beijing's Chipmaking Offensive.** Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu
2 **Investors Guide July 29, 2026.** Advantest, 30 July 2026, p. 12. advantest.com
3 **Testimony of David Isaacs, Semiconductor Industry Association, hearing on...** U.S. Senate Committee on Finance, 14 May 2025, p. 2. finance.senate.gov
4 **Investors Guide July 29, 2026.** Advantest, 30 July 2026, p. 10. advantest.com

earlier stages of chipmaking make dies, the individual chips on a wafer. The back end of the industry, assembly and test, checks which of them work and puts the good ones into packages.

Chokepoint card Test and Assembly, stage 12 of 14

Concentration

High

Four to six firms could supply this stage today, and the top two take most of the revenue.

Substitutability

Moderate

Rated on test tools, where Advantest held 62.7 percent of the market in 2025 and Teradyne 21.8 percent, by CSET's analysis of TechInsights data[5].

Price or market size

About $9.0 billion

market for chip testers in 2025, by Advantest's estimate[6]

Who leads	JP	**Advantest**	About 65% of the tester market in 2025, by its own estimate[6]
	US	**Teradyne**	$2.52bn of semiconductor test revenue in 2025, service included[7]
	TW	**ASE**	Ran 8,348 testers in Q2 2026, up from 6,797 a year earlier[8]
	US	**Amkor**	$6.71bn of 2025 revenue[9]
	JP	**Disco**	87% of the $1.9bn market for dicing saws and grinders in 2025, on CSET's analysis of TechInsights data[10]
Where it is made	TW	**Taiwan**	Taiwanese firms such as ASE and Powertech lead outsourced assembly and test
	JP	**Japan**	Advantest testers, Accretech and Tokyo Electron probers, Disco and Accretech dicing and grinding
	US	**United States**	Test equipment from Teradyne, FormFactor, Cohu and Aehr, and Amkor's planned Arizona packaging plant
	CN	**China**	Four of the top ten assembly and test firms by 2024 revenue, in a TrendForce ranking cited by CSET
	MY	**Malaysia**	Malaysia said in July 2026 that it ships 13% of the world's packaged chips
Why substitution is possible	Customers such as ASE already buy testers from both firms[11]. The barrier is cost, because switching means rebuilding a customer's whole test setup, Advantest says[2].		

2 Investors Guide July 29, 2026. Advantest, 30 July 2026, p. 12. advantest.com

5 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 20 October 2026. chipexplorer.eto.tech

6 Advantest, presentation notes for the FY2025 (year ended 31 March 2026) results... Advantest, 27 April 2026, p. 13. advantest.com

7 TERADYNE, INC, Form 10-K annual report for the period ended 2025-12-31 (10-K). U.S. Securities and Exchange Commission, 19 February 2026. sec.gov

8 ASE Technology Holding Co., Ltd., Form 6-K report of foreign private issuer for... U.S. Securities and Exchange Commission, 30 July 2026. sec.gov

9 AMKOR TECHNOLOGY, INC., Form 8-K current report for the period ended 2026-02-09... U.S. Securities and Exchange Commission, 9 February 2026. sec.gov

10 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

11 ASE Technology Holding Co., Ltd., Form 20-F annual report for the period ended... U.S. Securities and Exchange Commission, 1 April 2026. sec.gov

Where China stands

Georgetown's CSET, analyzing TechInsights data, finds that between 2019 and 2024 Chinese toolmakers gained 5 percentage points of world market share in test tools and 4 points in fabrication tools built for advanced packaging [1]. But they lost share in bonding and other conventional assembly and packaging tools [1].

Where the US stands

American firms held 23.5 percent of the market for test tools in 2025 and Japanese firms 63.0 percent, by CSET's analysis of TechInsights data [5]. Amkor raised its planned investment in an Arizona packaging and test campus to $7 billion in October 2025 [12]. In June 2026 it announced a long-term partnership with TSMC, which will buy packaging and test services from it in Arizona [13].

01

How it works

Figure 12.1 · Explainer, 23 seconds

How a chip is tested



1. **The wafer.** Earlier stages make the chips. Before any of them can be sold, a machine tests each one, starting while the chips are still on the wafer.
2. **Needles.** Thousands of needles on a probe card press onto one chip. They touch small metal pads on its surface.
3. **Questions and answers.** Through the needles, the tester sends signals into the chip and reads what comes back. It checks every answer.
4. **Chip by chip.** The wafer moves under the needles, chip by chip. The tester marks any chip that answers wrong, and that chip goes no further.
5. **Into the package.** Only chips that pass go on to be packaged. A modern accelerator bonds together a dozen expensive pieces, and one bad piece throws away all of them, so each piece is tested before it goes in.

Note: Simplified. Nine needles stand for thousands. A real prober, the machine that moves the wafer under the card, holds the card still and raises the wafer to meet the needles. The failed chips are placed for illustration and do not reflect a real yield. The package follows chapter 10: two logic chips and eight memory stacks on an interposer and a package base.

1 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu

5 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 20 October 2026. chipexplorer.eto.tech

12 Amkor Technology Breaks Ground on New Semiconductor Advanced Packaging and Test... Amkor Technology, 6 October 2025. ir.amkor.com

13 TSMC and Amkor Technology Announce Long Term Partnership to Accelerate Advanced... Amkor Technology, 16 June 2026. ir.amkor.com

Each round of test catches failures the round before it cannot see.

- **Wafer sort.** Thousands of needles land on each die while it is still on the wafer. A tester sends in test patterns and checks the answers, and dies that fail go no further.
- **Burn-in.** The part runs hot and at raised voltage for minutes to days, on the wafer or in the package, so that early-life failures happen in the factory, before the part ships[14].
- **Final test.** The tester checks the packaged part again, because packaging adds failures of its own.
- **System-level test.** The finished part sits in a socket and runs something close to a real workload, because some failures appear only when everything runs at once.

As the parts in a package grow more valuable, finding a bad die before packaging matters more[15]. Aehr, an American maker of burn-in equipment, sells wafer-level burn-in systems that look for early-life and hidden failures before dies are cut from the wafer[15]. Catching those failures early lowers the risk of packaging high-bandwidth memory and an expensive substrate with a processor that may later fail[15]. Aehr says its lead AI customer is moving all burn-in of that customer's AI accelerators from the system level, after packaging, to the wafer[16]. The FOX-XP systems that customer ordered from Aehr in August 2026 each burn in nine 300 mm wafers at a time[15].

02

Variants and trade-offs

Wafer sort and known-good die

The hard part of wafer sort is the probe card, the circuit board that holds the needles. For a large logic die every needle must sit in the same plane, survive repeated touchdowns without drifting, and do it hot. Each card is built for one chip design, so every new design needs a new card and a new qualification. FormFactor, an American probe card maker, says its cards work at high enough frequencies to compete well in testing high-bandwidth memory, a stack of eight, twelve or sixteen memory dies[17].

Final test, burn-in and system-level test

The bigger the package, the more one bad die can cost. TSMC told investors in July 2026 that its CoWoS roadmap includes packages larger than 14 times the area a lithography machine can print in one shot[18]. A die that fails after assembly wastes the memory and substrate joined to it[15].

14 AEHR TEST SYSTEMS, Form 10-K annual report for the period ended 2026-05-29. U.S. Securities and Exchange Commission, 27 July 2026. sec.gov

15 Aehr Receives $22 Million Follow-On Order for AI Processor Wafer-Level Burn-In... Aehr Test Systems, 12 August 2026. aehr.com

16 Aehr Test Systems Reports Fiscal 2026 Fourth Quarter and Full Year Financial... Aehr Test Systems, 14 July 2026. aehr.com

17 FORMFACTOR INC, Form 10-K annual report for the period ended 2025-12-27 (10-K). U.S. Securities and Exchange Commission, 20 February 2026. sec.gov

18 Q2 2026 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... LSEG StreetEvents, 16 July 2026, p. 16. investor.tsmc.com

Figure 12.2

Advantest put its tester market share at about 65 percent in 2025

Automated test equipment: share of the ~$9.0bn tester market, 2025



Source: Advantest FY2025 results briefing, April 2026 [6]

03

Who makes it

Advantest puts its own share at about 65 percent of an estimated $9.0 billion tester market in 2025[6]. Its share of system-on-chip (SoC) testers, the machines that test processors and other non-memory chips, rose about ten points in a year to 66 percent[6]. It held about 60 percent of memory test[6].

Teradyne, Advantest's main rival, took $2.52 billion of semiconductor test revenue in 2025, service included[7]. Of that, $1.89 billion came from SoC testers and $505 million from memory testers[7]. Teradyne's total revenue grew 13 percent to $3.19 billion in 2025, with fourth-quarter revenue up 44 percent on AI demand[19].

Georgetown University's Center for Security and Emerging Technology (CSET) makes a second estimate, from data by the research firm TechInsights[5]. It gives Advantest 62.7 percent of a $9.1 billion market for test tools in 2025 and Teradyne 21.8 percent[5]. Japanese firms held 63.0 percent of that market and American firms 23.5 percent[5].

Figure 12.3

SoC testers took an estimated $6.9 billion in 2025, memory $2.1 billion

Where the tester money went in calendar 2025



Source: Advantest FY2025 results briefing, April 2026 [6]

The rest of the test and assembly equipment market splits into niches, each with its own suppliers.

- **Probe cards.** FormFactor took $785.0 million of revenue in fiscal 2025, most of it from probe cards[17]. Technoprobe's revenue grew 15.7 percent to EUR 628.4 million in 2025, about 38 percent of it from AI[20]. It aims to double its manufacturing capacity by the end of 2027[20].

5 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 20 October 2026. chipexplorer.eto.tech

6 Advantest, presentation notes for the FY2025 (year ended 31 March 2026) results... Advantest, 27 April 2026, p. 13. advantest.com

7 TERADYNE, INC, Form 10-K annual report for the period ended 2025-12-31 (10-K). U.S. Securities and Exchange Commission, 19 February 2026. sec.gov

17 FORMFACTOR INC, Form 10-K annual report for the period ended 2025-12-27 (10-K). U.S. Securities and Exchange Commission, 20 February 2026. sec.gov

19 Teradyne Reports Fourth Quarter and Full Year 2025 Results. Teradyne, 2 February 2026. investors.teradyne.com

20 Technoprobe, press release: the Board of Directors approves the draft statutory... Technoprobe, 18 March 2026, p. 1. technoprobe.com

- **Probers**, the machines that move each wafer under the probe card and keep it at a set temperature. Tokyo Seimitsu, which sells under the Accretech name, says its own research gives it the largest global share in probers[21]. Tokyo Electron makes probers too[22]. CSET's analysis of TechInsights data counts handlers and probers as one market, worth $2.9 billion in 2025, and gives Tokyo Electron 19.9 percent of it and Accretech 17.1 percent[23].
- **Handlers and burn-in**, the machines that feed packaged parts into a tester and hold them at a set temperature, and the systems that run burn-in. Cohu took $453.0 million of revenue in fiscal 2025[24]. Aehr's revenue was $50.0 million in fiscal 2026, but it booked a record $60.7 million of orders in the fourth quarter alone[16]. It expects revenue of $130 million to $150 million in fiscal 2027[16].
- **Dicing and grinding.** Dicing saws cut wafers into dies, and grinders thin them. CSET's analysis of TechInsights data found that in 2025 Disco held 87 percent and Accretech 8.1 percent of the $1.9 billion market for these machines[10]. Panel-level packaging builds packages on large panels in place of 300 mm wafers, to make more of them at once[25]. In December 2025 Disco announced a saw that cuts such panels, as large as 400 by 400 mm, into individual packages[25].
- **Bonding.** Kulicke and Soffa took $654.1 million of revenue in fiscal 2025[26]. Besi's second-quarter 2026 revenue rose 68.7 percent to EUR 249.9 million on hybrid bonding and data center demand[27].

Thermocompression bonders use heat and force to press memory dies into a stack and to attach chiplets to substrates. ASMPT says it remained the leader in that market in 2025 and aims for a 35 to 40 percent share[28]. In its 2025 results it said it expects the market to reach $1.6 billion by 2028[28]. In July 2026 ASMPT announced new bulk orders for more than 50 of its chip-to-substrate bonders for AI chips, from outsourced assembly and test firms[29]. Hanmi Semiconductor in South Korea also sells thermocompression bonders for high-bandwidth memory[30]. See Memory and HBM.

04

The back end's geography

Taiwanese firms lead outsourced assembly and test, the OSAT business, and Chinese firms are large challengers. Taiwan's ASE billed $20,573.4 million in 2025[11]. Packaging made up about 48 percent of that, testing about 11 percent and its electronics

10 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

11 ASE Technology Holding Co., Ltd., Form 20-F annual report for the period ended... U.S. Securities and Exchange Commission, 1 April 2026. sec.gov

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21 Tokyo Seimitsu Integrated Report 2025. Accretech, 3 December 2025, p. 6. accretech.com

22 Precio™ Series. Tokyo Electron, accessed 9 September 2026. tel.com

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24 COHU, INC., Form 10-K annual report for the period ended 2025-12-27 (10-K). U.S. Securities and Exchange Commission, 17 February 2026. sec.gov

25 DISCO develops the DFD6080 fully automatic dicing saw for work sizes up to 400... DISCO Corporation, 15 December 2025. disco.co.jp

26 KULICKE AND SOFFA INDUSTRIES, INC., Form 8-K, exhibit 99.1: results for the... U.S. Securities and Exchange Commission, 19 November 2025. sec.gov

27 BE Semiconductor Industries N.V. Announces Q2-26 and H1-26 Results. BE Semiconductor Industries (Besi), 23 July 2026. besi.com

28 ASMPT Announces 2025 Annual Results AI-Driven Structural Growth Underpins Group... ASMPT, 4 March 2026. asmpt.com

29 ASMPT Secures Significant Wins for More than 50 Chip-to-Substrate... ASMPT, 29 July 2026. asmpt.com

30 Hanmi Semiconductor, products page. Hanmi Semiconductor, accessed 9 September 2026. hanmisemi.com

manufacturing arm about 40 percent[31]. In a TrendForce ranking that CSET cites, four of the top ten OSATs by 2024 revenue are headquartered in China[1].

Malaysia is a large center for chip assembly and test. Its investment agency said in July 2026 that the country ships 13 percent of the world's packaged chips[32]. It said a consortium of five Malaysian firms aims to help the country capture 7 percent of world advanced packaging shipments by 2035[32]. A 2024 study by the Semiconductor Industry Association and Boston Consulting Group used a different measure, capacity[33]. It gave Malaysia 7 percent of global assembly, test and packaging capacity in 2022, the largest share in Southeast Asia[33]. The study measured capacity by factory floor space at contractors and chipmakers' own plants[34].

Taiwan's assembly and test contractors are expanding. ASE ran 6,797 testers in the second quarter of 2025 and 8,348 a year later[8]. It spent $2.7 billion on machinery and $1.4 billion on buildings, facilities and automation in the first half of 2026[35]. In July 2026 it said revenue from leading-edge advanced packaging was running ahead of its $3.5 billion forecast for 2026[35]. Powertech, another Taiwanese assembly and test firm, aims to run the first panel-level packaging line for AI chips in 2027[36].

Amkor, an American assembly and test contractor, took $6.71 billion of revenue in 2025[9]. Its revenue in the second quarter of 2026 alone was $1.90 billion, up 26 percent year on year[37]. By October 2025 it had raised the planned investment in its Arizona campus to $7 billion[12].

In October 2024 TSMC and Amkor signed a memorandum of understanding for TSMC to buy packaging and test services from Amkor's planned plant in Peoria, Arizona[38]. TSMC would use those services for its customers, especially those using its Phoenix fabs[38]. The two announced a long-term partnership in June 2026[13].

1 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu

8 ASE Technology Holding Co., Ltd., Form 6-K report of foreign private issuer for... U.S. Securities and Exchange Commission, 30 July 2026. sec.gov

9 AMKOR TECHNOLOGY, INC., Form 8-K current report for the period ended 2026-02-09... U.S. Securities and Exchange Commission, 9 February 2026. sec.gov

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36 Powertech eyes world's first panel-level packaging for AI chips in 2027. Nikkei Asia, 27 August 2026. asia.nikkei.com

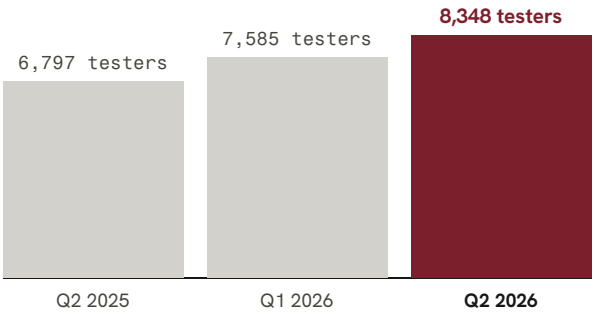
37 AMKOR TECHNOLOGY, INC., Form 8-K current report for the period ended 2026-07-27... U.S. Securities and Exchange Commission, 27 July 2026. sec.gov

38 Amkor and TSMC to Expand Partnership and Collaborate on Advanced Packaging in... TSMC, 4 October 2024. pr.tsmc.com

Figure 12.4

ASE ran 8,348 testers in mid-2026, up from 6,797 a year earlier

Testers installed at ASE, a Taiwanese assembly and test contractor



Source: ASE Technology Holding, second quarter 2026 earnings release (SEC Form 6-K) [8]

05

The chokepoint

Customers can switch tester makers, but at a cost. Advantest says that switching to another tester maker means rebuilding the whole test setup used for development, evaluation and mass production, at significant extra cost to the customer[2]. Still, ASE says it buys testers with similar functions from different makers, including Teradyne and Advantest[11].

Chinese toolmakers are gaining share in test and advanced packaging tools, but CSET finds they still lag significantly behind foreign suppliers in market share[1]. Between 2019 and 2024 they gained 5 percentage points of world market share in test tools and 4 points in fabrication tools built for advanced packaging, by CSET's analysis of TechInsights data[1]. But they lost share in bonding and other conventional assembly and packaging tools[1].

Assembly and test contractors such as ASE and Amkor earn a small part of what the chip industry sells. ASE's packaging business took $9,829.2 million from outside customers in 2025, and its testing business $2,292.0 million[11]. Amkor's revenue was $6.71 billion[9]. Global semiconductor sales that year were $791.7 billion[39].

Three questions to answer from memory

1. Why do chipmakers test chips early and often? see How it works
2. Who makes most chip testers? see Who makes it

1 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu

2 Investors Guide July 29, 2026. Advantest, 30 July 2026, p. 12. advantest.com

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39 Global Annual Semiconductor Sales Increase 25.6% to $791.7 Billion in 2025. Semiconductor Industry Association, 6 February 2026. semiconductors.org

3. How many of the ten largest chip assembly and test contractors are based in China?
   see The back end's geography

Three things to remember

- Chips are tested early and often because one bad chip can ruin a package of expensive ones.
- Advantest and Teradyne account for most spending on chip testers.
- In a TrendForce ranking cited by CSET, four of the top ten outsourced assembly and test firms by 2024 revenue are based in China.

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A company filings, governments, standards bodies, universities and research institutions; **B** SemiAnalysis, Reuters, the Financial Times and Nikkei Asia.

How to cite this chapter

> Sharma, Yashvardhan. “Test and Assembly.” *Chip Supply Chain: How AI Chips Are Made*. Edition of 28 October 2026. https://chipsupplychain.org/test-and-assembly/

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@misc{chipsupplychain2026testandassembly,
  title        = {Test and Assembly},
  author       = {Sharma, Yashvardhan},
  howpublished = {Chip Supply Chain: How AI Chips Are Made},
  year         = {2026},
  month        = {October},
  note         = {Edition of 28 October 2026},
  url          = {https://chipsupplychain.org/test-and-assembly/}
}
```